

Recipe practice solutions

Feedforward transformations

Open a solution only after attempting that question. Each green box is the result.

1 One pass through all four steps

1. The input currents are linear in the represented scalar:

$$J_1^A = x + 1, \quad J_2^A = -x + 1.$$

The output currents are the same kind of encoding, written in y rather than x :

$$J_1^B = 2y + \frac{1}{2}, \quad J_2^B = -y + 1.$$

The activities of A are already given, so these input currents are not turned back into rates.

$$J_1^A = x + 1, \quad J_2^A = -x + 1. \quad J_1^B = 2y + 1/2, \quad J_2^B = -y + 1.$$

2. The value represented by B is twice the value represented by A:

$$y = 2x.$$

$$y = 2x.$$

3. Rows are neurons and columns are the inputs $x = 0$ and $x = 1$:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 2 \end{pmatrix}.$$

$$AA^T = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}, \quad (AA^T)^{-1} = \frac{1}{9} \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}, \quad AY^T = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

$$(D^f)^T = \frac{1}{9} \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 4/3 \\ -2/3 \end{pmatrix}, \quad D^f = \begin{pmatrix} 4/3 & -2/3 \end{pmatrix}.$$

On the two samples, $a = (1, 2)$ and $a = (2, 1)$:

$$D^f a = 0 \quad \text{and} \quad D^f a = 2,$$

which are $2x$ at $x = 0$ and $x = 1$. Two neurons and two samples, with this invertible A , can hit both targets exactly.

$$D^f = (4/3, -2/3). \text{ It reproduces } (0, 2).$$

4. The output encoders, with gains included, are the rows of

$$E = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

$$W^f = ED^f = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \begin{pmatrix} 4/3 & -2/3 \end{pmatrix} = \begin{pmatrix} 8/3 & -4/3 \\ -4/3 & 2/3 \end{pmatrix}.$$

Biases stay outside the matrix:

$$J_1^B = \frac{8}{3}a_1 - \frac{4}{3}a_2 + \frac{1}{2}, \quad J_2^B = -\frac{4}{3}a_1 + \frac{2}{3}a_2 + 1.$$

At $x = 1$ the activities are $a_1 = 2$, $a_2 = 1$, and $2x = 2$. From the weights,

$$J_1^B = \frac{8}{3} \cdot 2 - \frac{4}{3} \cdot 1 + \frac{1}{2} = \frac{9}{2}, \quad J_2^B = -\frac{4}{3} \cdot 2 + \frac{2}{3} \cdot 1 + 1 = -1.$$

From the output encoding with y replaced by $2x$,

$$J_1^B = 2 \cdot 2 + \frac{1}{2} = \frac{9}{2}, \quad J_2^B = -2 + 1 = -1.$$

Same currents. That is the substitution the fourth step asks for.

$$W^f = \begin{pmatrix} 8/3 & -4/3 \\ -4/3 & 2/3 \end{pmatrix}. \text{ At } x = 1 \text{ both routes give } J = (9/2, -1).$$

2 Same populations, new function

1. The encodings of A and of B were fixed by the populations, so step 1 is unchanged. Step 2 is a new sentence, $y = x + 1$. Steps 3 and 4 use that new target, so Y , D^f , and W^f are recomputed. E and the biases belong to B and stay put.

Copy step 1, and copy E . Redo the transformation, D^f , and W^f .

2.

$$Y = \begin{pmatrix} 1 & 2 \end{pmatrix}, \quad AY^T = \begin{pmatrix} 5 \\ 4 \end{pmatrix}.$$

$$(D^f)^T = \frac{1}{9} \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad D^f = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

The second input neuron receives a decoder of zero for this particular target. That is a fact about these samples, not a rule about constant offsets.

$$D^f = (1, 0).$$

3.

$$W^f = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & 0 \end{pmatrix}.$$

$$J_1^B = 2a_1 + \frac{1}{2}, \quad J_2^B = -a_1 + 1.$$

At $x = 1$, $a_1 = 2$ and $y = 2$, so $J_1^B = 9/2$ and $J_2^B = -1$, which is $\alpha ey + J^{\text{bias}}$ for each neuron of B.

$$J_1^B = 2a_1 + 1/2, \quad J_2^B = -a_1 + 1.$$

4.

$$d + d^1 = \left(\frac{2}{3}, -\frac{1}{3}\right) + \left(\frac{1}{3}, \frac{1}{3}\right) = (1, 0),$$

which is the decoder from part 2. Least squares is linear in the target, and $x + 1$ is the identity plus the constant function. The identity decoder alone reconstructs x . Substituting that into B would encode x , and the $+1$ would be missing.

$$d + d^1 = (1, 0). \text{ The identity decoder implements } y = x.$$